

## Multiple Choice

1. **Answer:** A

*Options:* A) 9; B) 37; C) -23; D) 14; E) -14

*Note:* Correct option: A - 9

2. **Answer:** A

*Options:* A) 0; B) 2; C) 6; D) 80; E) 20

*Note:* Correct option: A - 0

3. **Answer:** A

*Options:* A) 8; B) 10; C) 4; D) 14; E) 6

*Note:* Correct option: A - 8

4. **Answer:** A

*Options:* A)  $E(X + Y) = 307$ ,  $\text{var}(X + Y) = 10$ ; B)  $E(X + Y) = 307$ ,  $\text{var}(X + Y) = 6$ ; C)  $E(X + Y) = 619$ ,  $\text{var}(X + Y) = 14$ ; D)  $E(X + Y) = 619$ ,  $\text{var}(X + Y) = 7$ ; E)  $E(X + Y) = 312$ ,  $\text{var}(X + Y) = 14$

*Note:* Correct option: A -  $E(X + Y) = 307$ ,  $\text{var}(X + Y) = 10$

5. **Answer:** A

*Options:* A) Sunday; B) Wednesday; C) Tuesday; D) Saturday; E) The day of the week cannot be determined

*Note:* Correct option: A - Sunday

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'False, Not Enough Information'.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '0'.

8. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '-2'.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** 2. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 3. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*